DOUGLAS COUNTY GOVERNMENT

Finance Department ~ Purchasing Division
100 Third Street, Suite 130
Castle Rock, Colorado 80104
Telephone: 303-663-6161
www.douglasco.gov

INVITATION FOR BID (IFB)
NO. 042-26
HVAC FILTERS & BELTS

YOUR BID RESPONSE MUST BE RECEIVED NO LATER THAN
NOVEMBER 18, 2026 3:00 PM

BIDDER'S CERTIFICATION

We offer to provide to Douglas County the materials, supplies, products and/or services requested in accordance with the specifications and subject to the terms and conditions of the purchase(s) described herein:

BIDDER: _____

ADDRESS: _____

CITY: _____ **STATE:** _____ **ZIP:** _____

TELEPHONE NUMBER: _____ **FAX NUMBER:** _____

EMAIL: _____

BY: _____
(Printed or Typed Name)

_____
(Signature)

TAXPAYER I.D. NUMBER: _____

Signature constitutes acceptance of all terms and conditions listed on this form and all documents attached.

Bids shall be received **ELECTRONICALLY ONLY** through the Rocky Mountain E-Purchasing/BidNet System website at (www.bidnetdirect.com/colorado/douglas-county-government). All bid responses should consist of one (1) PDF document, without password protection. It is the sole responsibility of the bidder to ensure that their bid response is received on time. Bids will not be considered which are received after the time stated.

Douglas County distributes solicitation documents through the Rocky Mountain E-Purchasing/BidNet System only. Copies of solicitations obtained from other sources are not considered official copies and the County cannot attest to their accuracy. All BidNet System questions must be addressed with BidNet Vendor Support at 800-835-4603.

Douglas County Government reserves the right to reject any or all bids, to waive formalities, informalities, or irregularities contained in the said bid and furthermore, to award an agreement/contract for items herein, either in whole or in part, if it is deemed to be in the best interest of the County to do so. Additionally, the County reserves the right to negotiate optional items and/or services with the successful bidder.

OVERVIEW:

Through this Invitation for Bid (IFB), the Department of the Facilities & Fleet Services of Douglas County Government, hereinafter referred to as the County, respectfully requests bids from responsible and highly-qualified companies for the provision of the purchase of HVAC Filters and Belts for use within Douglas County Facilities, as specified.

The initial agreement/contract, issued as a result of this IFB, will be for a period of twelve (12) months, beginning on or about January 1, 2027, and ending on December 31, 2027. All pricing shall be firm and fixed for the initial agreement/contract period. The County shall have the option to renew the award for four (4) additional periods of one (1) year each, with the same terms and conditions. This agreement/contract and any extension to the original period of a subject agreement/contract shall be contingent upon annual funding being appropriated, budgeted, and otherwise made available for such purposes and subject to the County's unanimous satisfaction with the products and services received during the preceding agreement/contract period.

BID CALENDAR:

October 23, 2026	Invitation for Bid (IFB), posted on the RME-Purchasing System
November 2, 2026	Invitation for Bid (IFB), Deadline for submission of questions (3:00 pm)
November 9, 2026	Invitation for Bid (IFB), Posting of questions/responses
November 18, 2026	Invitation for Bid (IFB), Due Date (see page one)
December 2026	Interview of Finalists, if necessary
January 2027	Contract Start Date

BID CONDITIONS AND PROVISIONS:

All bids must be submitted on the forms provided and in accordance with all terms, conditions, specifications, and stipulations contained herein. Bidders shall carefully read and be familiar with all terms, conditions, specifications, and stipulations contained in this IFB, which shall become part of the final agreement/contract.

All bids must be signed by a duly-authorized official of the company. The completed and signed bidder certification page (together with all required attachments) must be uploaded to the Rocky Mountain E-Purchasing/BidNet System before the due date and time shown on page one (1).

All participating bidders, by their signature on the certification page, shall agree to comply with all terms, conditions, specifications, and stipulations of this IFB as stated or implied herein. Any alteration, erasure or interlineation by the bidder in this IFB shall constitute cause for rejection by the County. Exceptions or deviations to this IFB must not be added to the IFB pages and must be on company letterhead and accompany the bid response.

All bidders are required to complete all of the information requested in this IFB. Failure to do so may result in the disqualification of your bid response. When omitting a bid on an item, please insert the words "NO BID".

Prices stated must be in the units specified. In the case of a discrepancy between the unit price and extension price, the unit price shall be considered correct. Minor details omitted by oversight will not constitute an excuse for their omission.

Bid pricing shall be fixed and subject to acceptance or rejection within ninety (90) days of the bid due date.

The County will not be responsible for any goods delivered and/or services performed without its purchase order, signed by an authorized representative of the County's Purchasing Division.

The successful bidder will be required to provide proof of and the required limits of liability insurance, including Workers' Compensation. This proof of insurance must be in the form of a "Certificate of Insurance" and must show coverage in the amounts specified by the laws of the State of Colorado for the duration of any agreement/contract issued as a result of this IFB. Additionally, the County must be notified of any changes

occurring in this coverage and the successful bidder must demonstrate to the County that such changes do not, in any way, affect the minimum liability insurance required for this bid.

All information submitted in response to this IFB may be subject to disclosure under the Colorado Open Records Act. Bidders are discouraged from providing information that they consider confidential and/or privileged as part of their response to this IFB. If any portion of your response is identified as proprietary and/or confidential, a redacted copy must be uploaded to the Rocky Mountain E-Purchasing/BidNet System with your original/complete response.

The successful bidder may be required, as a provision of this IFB, to submit proof of compliance with governmental health and safety codes, regulations and standards, as appropriate.

The successful bidder shall be in complete compliance with all terms, conditions, specifications, and stipulations of this IFB as outlined herein. The County shall have the right to inspect the facilities and equipment of the successful bidder to ensure such compliance.

No bid shall be accepted from, and no agreement/contract will be awarded to, any person, firm or corporation that is in arrears to the County, upon debt or contract, or that is a defaulter, as surety or otherwise, upon any obligation to the County, or that has failed to attain or demonstrate compliance with any law, ordinance, regulation, or agreement/contract term or condition as may be provided for or required in any County agreement/contract, or that may be deemed irresponsible or unreliable by the County. Bidders may be required to submit satisfactory evidence that they have a practical knowledge of the particular work bid upon and that they have the necessary financial resources to perform and complete the work outlined in this IFB.

All bidders agree to abide by all of the laws, regulations, and administrative rulings of the United States, the State of Colorado, and the County of Douglas, securing all necessary licenses/permits in connection with this IFB.

All materials furnished or services performed under the terms of this agreement/contract issued as a result of this IFB shall comply with the requirements and standards specified in the Williams-Steiger Occupational Safety and Health Act (OSHA) of 1970 (Public Law 91-596), as well as with other applicable federal, state, and local codes.

In the event that this IFB requires a formal contract to be prepared by the County, the successful bidder will properly sign and furnish any bonds, insurance, Workers' Compensation, etc. as may be required by the County within ten (10) days (unless a longer period is allowed by the County) from the date of receipt of the formal contract forms.

All bidders must take into consideration that only the County's documents will be used in the finalization of any contract issued as a result of this IFB. Bidders are responsible for reviewing such documents prior to submitting their bid response.

All bidders must submit written disclosure of any known potential conflicts of interest that may result during the purchase of goods and/or the performance of the services listed herein.

ETHICAL STANDARDS:

It shall be a breach of ethical standards for any person to offer, give, or agree to give any employee or former employee, or for any employee or former employee to solicit, demand, accept, or agree to accept from another person, a gratuity or an offer of employment in connection with any decision, approval, disapproval, recommendation, preparation of any part of a program requirement or purchase request, influencing the content of any specification or procurement standard, rendering of advice, investigation, auditing, or in any other capacity in any proceeding of application, request for ruling, determination, claim or controversy, or other particular matter, pertaining to any program requirement or an agreement/contract or subcontract, or to any solicitation or IFB, therefore.

It shall be a breach of ethical standards for any payment, gratuity, or offer of employment to be made by or on behalf of a subcontractor under an agreement/contract to the prime contractor or higher tier subcontractor or any person associated therewith, as an inducement for the award of a subcontract or order.

In the event that any gratuities or kickbacks are offered or tendered to any County employee or representative, the bid response shall be disqualified and shall not be reinstated.

ETHICSPOINT/NAVEX GLOBAL:

Douglas County is committed to an environment where open, honest communication is the expectation, not the exception. Douglas County has implemented a phone and Internet-based reporting system called EthicsPoint/Navex Global. The reporting system allows employees to report issues or concerns that they may have in an anonymous and confidential manner. We consider it a critical component in providing a safe, secure, and ethical workplace.

An option within the system allows vendors to also report an issue. As a vendor, you are in a position to observe not only questionable or unethical behavior by our employees but also identify areas that you would like to make a suggestion for change. By calling 1-888-337-3094 or logging-in on the Internet to www.ethicspoint.com and entering Douglas County Government, you can file a report, offer a suggestion or voice a concern. Online instructions are available on how to use the system. The EthicsPoint/Navex Global system also offers a follow-up/response mechanism. You can be assured that this is a fast, easy, and confidential system.

NON-COLLUSIVE BIDDING CERTIFICATION:

By the submission of your bid response, the bidder certifies that:

1) The bid has been arrived at by the bidder independently and has been submitted without collusion with any other bidder.

2) The contents of the bid have not been communicated by the bidder; nor, to the best of their knowledge and belief, by any of their employees or agents, to any person not an employee or agent of the bidder or the surety on any bond furnished herewith, and will not be communicated to any such person prior to the official opening of this bid.

3) No bidder shall submit more than one (1) bid for this purchase.

NON-DISCRIMINATION IN EMPLOYMENT:

In connection with the performance of the work under an agreement/contract issued as a result of this IFB, the successful bidder must agree not to refuse to hire, discharge, promote or demote, or to discriminate in matters of compensation against any person otherwise qualified, solely because of race, color, religion, national origin, gender, age, military status, sexual orientation, marital status, or physical or mental disability. The successful bidder must further agree to insert the foregoing provision into all subcontracts.

INDEMNIFICATION:

The County cannot and by this agreement/contract does not agree to indemnify, hold harmless, exonerate or assume the defense of the company or any other person or entity whatsoever, for any purpose whatsoever. The company shall defend, indemnify and hold harmless the County, its commissioners, officials, officers, directors, agents, and employees from any and all claims, demands, suits, actions or proceedings of any kind or nature whatsoever, including Workers' Compensation claims, in any way resulting from or arising from the services rendered under this agreement/contract; provided, however, that the company need not indemnify or save harmless the County, its officers, agents and employees from damages resulting from the sole negligence of the County's commissioners, officials, officers, directors, agents, and employees.

INDEPENDENT CONTRACTOR:

The company is an Independent Contractor and is free to perform services for other clients. Notwithstanding any provision of this agreement/contract, all personnel assigned by the company to perform work under this agreement/contract shall be, and remain at all times, employees of the company for all purposes. The County shall have no responsibility for any federal and state taxes and contributions for Social Security, unemployment insurance, income withholding tax, and other taxes measured by wages paid to employees of the company and/or its designated agents. The company acknowledges that it and its employees are not entitled to Workers' Compensation benefits or Unemployment Insurance benefits from the County, unless the company or a third party provides such coverage, and that the County does not pay for or otherwise provide such coverage. The company shall provide and keep in force Workers' Compensation (and provide proof of such insurance when requested by the County) and Unemployment Compensation insurance in the amounts required by law, and shall be solely responsible for its own actions, its employees and agents.

ASSIGNMENT OF PURCHASE ORDER OR CONTRACT:

The successful bidder may not assign or otherwise transfer any of its rights or obligations under any purchase order or agreement/contract made and entered into pursuant to the County IFB without the prior written approval of the County Purchasing Division.

CANCELLATION:

The County reserves the right to cancel the whole or any part of an agreement/contract issued as a result of this IFB due to the failure of the successful bidder to carry out any term, promise or condition of the agreement/contract. The County will issue a written notice of default to the successful bidder for failing to act in compliance with the terms and conditions of such an agreement/contract.

EXAMINATION OF RECORDS:

The successful bidder agrees that the County shall, until the expiration of three (3) years after the final payment under an agreement/contract issued as a result of this IFB, have access to and the right to examine any directly pertinent books, documents, papers, and records of the bidder involving transactions of such agreement/contract.

PRICE, PAYMENT, AND DELIVERY:

1) All prices shall be firm and fixed for the specified time frame.

2) All payment terms shall be "Net 30"; consideration will be given to any discounts offered for payments made earlier than "Net 30", please explain.

3) In an effort to improve efficiency, Douglas County encourages the use of purchasing cards to make payments. Under this method, payments are made after the verification of the receipt of the goods and services and following the receipt of a proper invoice. Please confirm if this would be an acceptable payment option.

4) All deliveries will be "F.O.B. ~ Destination"

TAXES:

All pricing shall reflect all applicable tax exemptions for Douglas County:

1) Federal Registration Number: 84-6000761

2) State Registration Number: 98-04428

COOPERATIVE PURCHASING:

Douglas County encourages cooperative purchasing in an effort to assist other agencies to reduce their cost of bidding and to make better use of taxpayer dollars through volume purchasing. Vendors may, at their discretion, agree to extend the prices and/or terms of the resulting award to other state or local government agencies, school districts, or political subdivisions in the event they would have a need for the same product/service. Usage by any entity shall not have a negative impact on Douglas County in the current term or in any future terms. The vendor must deal directly with any governmental agency concerning the placement of purchase orders/agreements, freight/delivery charges, contractual disputes, invoices, and payments. Douglas County shall not be liable for any costs or damages incurred by any other entity.

REFERENCE LISTING:

Bidders shall furnish a list of three (3) current references to include name, title, address, telephone number, and email address for which they are currently furnishing or have in the past sold products and/or furnished services on a same or similar agreement/contract. The inability to contact references may be cause for the rejection of your bid response.

QUESTIONS/INQUIRIES/ADDENDUMS:

All questions related to this IFB must be submitted **ELECTRONICALLY ONLY** through the Rocky Mountain E-Purchasing/BidNet System website at (www.bidnetdirect.com/colorado/douglas-county-government). Please enter only one (1) question per line.

Questions will be received until 3:00pm on November 2, 2026. Responses to all questions will be posted on the Rocky Mountain E-Purchasing/BidNet System no-later-than November 9, 2026.

If it becomes necessary to revise any part of this IFB or if additional data is necessary to enable an exact interpretation of the provisions of this request, an addendum will be issued. It is the responsibility of the potential bidder to ensure that they have received all addendums prior to submitting their IFB response.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS
VENDOR INSTRUCTIONS

GENERAL INFORMATION:

1) The Facilities & Fleet Services Department of Douglas County Government, hereinafter referred to as the County, respectfully requests bids from responsible qualified firms for the provision of the purchase of HVAC Filters and Belts for use within Douglas County buildings, as specified.

2) The County's intent is to award this IFB to only one (1) firm for the provision of the purchase of HVAC Filters and Belts; however, the results of this IFB do not obligate the County in any manner. The bid schedules are attached and incorporated herein; bid comparisons will be made from those schedules. Reference checks and past experience will also be considered in the evaluation process.

3) The contract placed with the successful bidder, as a result of this IFB, shall be a Blanket Purchase Agreement. The Blanket Purchase Agreement authorizes, but does not obligate, the County to order any minimum or maximum, dollar amount or quantity of filters and/or belts.

4) The successful bidder must be able to provide a single point of contact for all customer service requirements to include: ordering, scheduling deliveries, and/or requesting information.

5) The successful bidder must be able to meet specified delivery dates and times, to be determined following bid award. The County reserves the right to refuse any delivery that does not occur within the specified delivery dates and times unless special arrangements have been made.

6) All bids must include any and all applicable costs to include shipping and handling.

7) The successful bidder must have a stocked warehouse, within a seventy-five (75) mile radius of Castle Rock, Colorado, for all items listed within this IFB.

8) Bidders must explain, in detail, their company's return policy. **No "re-stocking charges" will be paid**.

9) The successful bidder will furnish to Douglas County Government, at no additional cost, any reports that may be needed in tracking the ordering and usage of said products within the County. These reports will also be used to verify contract pricing.

10) All bid responses should provide detailed information in regards to any cash incentives, rebates, products, and/or value-added services that are being offered and/or considered as a part of this potential award.

11) All bidders must specify whether or not they will accept a County MasterCard, without an additional fee, for payment of any or all invoices. In addition, provide the amount of a discount that would be applied to all base payment amounts if those base payment amounts were made via County purchasing card.

12) All bid responses should provide an explanation as to why your company would stand out above all others and why your company should be awarded the contract resulting from this IFB.

13) After reviewing submissions to this IFB, the County will contact selected vendors with the most complete bids and the lowest costs to request product samples. Samples will be required for each unique product design or differing specification, but not for every belt or filter size.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS
INSTRUCTIONS TO BIDDERS

BID SPECIFICATIONS FOR PURCHASE OF HVAC FILTERS and BELTS:

1) Activities to be Performed:
   Provide HVAC Filters and Belts

2) Delivery Locations:
   - Douglas County Operations Facility (TinTech), 3026 North Industrial Way, Castle Rock, Colorado, 80109
   - Robert A. Christensen Justice Center, 4000 Justice Way, Castle Rock, Colorado, 80109

3) Bid Inclusions:
   All bid responses should include all shipping, handling, and delivery charges.

4) Bid Exclusions:
   Douglas County Facilities Maintenance will perform all installation.

5) Warranty:
   An industry standard warranty of material and workmanship of not less than one (1) year from the date of acceptance by Douglas County.

6) References:
   Bidders shall furnish a list of three (3) current references to include name, title, address, telephone number, and email address for which they are currently furnishing or have in the past sold products on a same or similar contract or agreement. The inability to contact references may be cause for rejection of your bid response.

7) All orders must be packaged and labeled per facility and per equipment number (no bulk packaging).

8) All orders are to be set up for auto ship per Douglas County's Maintenance Schedule.

9) No minimum order requirements.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS
DETAILED BID SPECIFICATIONS

SECTION ONE
MERV-9 PLEATED PANEL FILTERS

1) SCOPE

This specification covers pleated panel filters that are a component of air-conditioning, heating, and ventilating systems.

2) PURPOSE

The purpose of this specification is to establish performance criteria and identify physical properties that are pertinent and necessary for proper filter performance. Conformance to all items in the specification is the responsibility of the bidder.

3) BID ENCLOSURES

The following items are required as part of the bid response.

3.1 A product sample for each distinctively different product design.

4) GENERAL

4.1 Air filters shall be medium efficiency ASHRAE pleated panels consisting of cotton and synthetic media, welded wire media support grid, and beverage board enclosing frame.

4.2 Sizes shall be noted on drawings or other supporting materials.

5) CONSTRUCTION

5.1 Filter media shall be two layers of polyester, lofted to a uniform depth of 0.11", and formed into a uniform radial pleat. Filter media shall have a MERV 9 efficiency after conditioning methods as stated in ASHRAE 52.2, Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size, Appendix J or in the discharging method designated in ISO 16890.

5.2 A welded wire grid of 0.020" in diameter, spot-welded on one-inch centers and treated for corrosion resistance shall be bonded to the downstream side of the media to maintain radial pleats and prevent media oscillation.

5.3 An enclosing frame of no less than 28-point high wet-strength beverage board shall provide a rigid and durable enclosure. The frame shall be bonded to the media on all sides to prevent air bypass. Integral diagonal support members on the air entering and air exiting side shall be bonded to the apex of each pleat to maintain uniform pleat spacing in varying airflows.

6) PERFORMANCE

6.1 The filter shall have a Minimum Efficiency Reporting Value of MERV 9 when evaluated under the guidelines of ASHRAE Standard 52.2. It shall have a MERV-A of 9 when tested per Appendix J of the same standard. The media shall maintain or increase in efficiency over the life of the filter.

6.2 The filter shall have a minimum efficiency rating of ISO ePM10 -55 when tested to ISO standard 16890.

6.3 Initial resistance to airflow shall not exceed 0.30" w.g. for 2" configurations at 500 fpm, and 0.27" w.g. for 4" configurations at 500 fpm.

6.4 The filter shall have a Minimum Efficiency Reporting Value of MERV 9 when evaluated under the guidelines of ASHRAE Standard 52.2, Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size, Appendix J. The media shall maintain or increase in efficiency over the life of the filter.

6.5 Initial resistance to airflow shall not exceed 0.32" w.g. for 2" configurations at 500 fpm, and 0.29" w.g. for 4" configurations at 500 fpm.

6.6 The filter shall have an Energy Cost Index (ECI) value of five stars.

6.7 The filter shall be classified by Underwriters Laboratories as UL Class 900.

6.8 Manufacturer shall provide evidence of facility certification to ISO 9001:2008.

6.9 Manufacturer shall guarantee the integrity of the filter pack to 2.0" w.g.

6.10 Manufacturer shall guarantee the efficiency level will always be at or exceed MERV 9 when tested according to ASHRAE 52.2, or ISO ePM_{10} 50% when tested to ISO 16890, throughout the entire service life of the filter.

6.11 Manufacturer shall guarantee the filter will last twelve (12) calendar months in applications operating at or less than 100 hours per week, before reaching 1.0" w.g. of static pressure across the filter OR the manufacturer shall guarantee the filter will last nine (9) calendar months in applications operating above 100 hours per week, before reaching 1.0" w.g. of static pressure across the filter. Both guarantees are to be valid at or below the rated air velocity of 500 fpm.

6.12 Supporting Data - Provide product test report including all details as prescribed in ASHRAE Standards 52.2, including Appendix J.

7) PACKAGING AND IDENTIFICATION

7.1 Air filter identification markings on each filter frame shall include
- 7.1.1 Filter Manufacture
- 7.1.2 Filter Type/Model Number
- 7.1.3 Nominal Filter Size/Actual Filter Size
- 7.1.4 MERV Rating
- 7.1.5 UL 900 Classification
- 7.1.6 Code Build Date

7.2 Filter packing carton markings shall include
- 7.2.1 Filter Manufacture
- 7.2.2 Filter Type/Model Number
- 7.2.3 Nominal Filter Size
- 7.2.4 MERV Rating
- 7.2.5 UL 900 Classification
- 7.2.6 Manufacture Part Number
- 7.2.7 Carton Quantity

7.3 Packaging shall provide adequate protection against damage or deterioration during shipment and allow complete identification of both the filters and the shipping container.

8) PERFORMANCE CRITERIA

8.1 The manufacturer shall have representative samples of the product requested in the bid solicitation tested by an Independent test laboratory approved by the purchaser. The test laboratory shall test one (1) representative filter, tested to ASHRAE standard 52.2.-2012 and with Appendix J, five (5) years or newer.

8.2 The resulting performance data shall be used to meet the following:

8.2.1 NOMINAL FILTER SIZE (W X H X D) -- 24x24x2
Gross Media Area -- 17.3

8.2.2 TEST CONDITIONS
Filter Face Velocity (CFM) -- 2000
Initial Clean Resistance (Inches W.G.) -- 0.32"
Final Dirty Resistance (Inches W.G.) -- 1.00"

8.2.3 PERFORMANCE REQUIREMENTS-ASHRAE 52.2-2012
Minimum Composite Average Efficiency (%) under Appendix J: E1: 8; E2: 39; E3: 80
Minimum Efficiency Reporting Value (MERV-A) 9

8.2.4 PERFORMANCE REQUIREMENTS-ASHRAE 52.-2012
Minimum Average Arrestance (%) -- 92
Measured dust holding capacity (DHC) -- 200 grams

8.3 The Supplier shall agree to have a representative make on-site inspections on a quarterly basis. The purpose of this inspection is to ensure that the filters are functioning properly.

9) BID ENCLOSURES

The following items are required as part of the bid response.

9.1 For each indicated filter size and capacity, one (1) laboratory test report showing performance results when tested to ASHRAE standard 52.2-2012. Permission shall be granted by the bidder to verbally verify the performance results with the laboratory that performed the tests.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS
DETAILED BID SPECIFICATIONS

SECTION TWO
RING PANEL FILTERS 2-PLY

1) SCOPE

This specification covers panel filters that are a component of air-conditioning, heating, and ventilating systems.

2) PURPOSE

The purpose of this specification is to establish performance criteria and identify physical properties that are pertinent and necessary for proper filter performance. Conformance to all items in the specification is the responsibility of the bidder.

The Supplier shall agree to have a representative make on-site inspections on a quarterly basis. The purpose of this inspection is to ensure that the filters are functioning properly.

3) PHYSICAL CHARACTERISTICS

The filter panel shall be of a two media layer design, securely heat sealed around an internal heavy wire grid. To prevent media fluttering and sifting of trapped dust, the media layers shall have 4 internal stabilizer seals totaling not less than 60 inches in total length.

4) FILTER MEDIA

A one-half inch layer of 15 denier totaling .39 ounces per square foot shall form the upstream media. A heavily needled 6 denier media, .46 ounces per square foot, shall form the downstream media. Total media weight shall be 7.6 ounces per square yard.

The panel shall have a non-migratory, non-drying adhesive between the two layers to maximize efficiency and dust retention.

The air filter shall contain an antimicrobial biocide. Biostats are not acceptable. The antimicrobial shall effectively control the growth of mold, algae, mildew, and fungi on the filter. The antimicrobial biocide shall be part of the binder that is applied during manufacturing of the media itself.

The filter shall not support the growth of microbes. The antimicrobial must have no water solubility. The antimicrobial must be registered by the E.P.A. for the use in filters. The antimicrobial must also have clearance for use under regulations of U.S.D.A. and F.D.A. All testing on the antimicrobial must be in compliance with the American Society of Test Methods (ASTM).

The filters shall yield the following typical performance when tested by ASHRAE TEST STANDARD 52.1-1992:

TEST FILTER	24x24x1	AVERAGE EFFICIENCY	28%
RESISTANCE at 300 FPM	0.13 w.g."	AVERAGE ARRESTANCE	86%
INITIAL EFFICIENCY	12%	DUST HOLDING CAPACITY @ 1.00 w.g."	198 grams

Filter shall be MERV 7 by ASHRAE 52.2 - 1999 Independent Laboratory test at 295 fpm.

Range 1 (3.0%)	Range 2 (23.0%)	Range 3 (59.1%)
(0.3 - 1.0 microns)	(1.0 - 3.0 microns)	(3.0 - 10.0 microns)

a) **WIRE FRAME**
The internal wire support frame shall be a 9-gauge wire frame strength or greater, with internal grid support.

b) **PANEL EDGES**
Panel edges shall have media outside the wire frame grid of 1" to 1.5" of overhang to assure self-sealing (no dirt bypass).

c) **CONTROL MANUFACTURING OF END PRODUCT**
In order to control the specified panel filter product the manufacturer of the final panel product shall also be the manufacturer of the media used to make the panel filter.

5) PACKAGING AND IDENTIFICATION

Packaging shall provide adequate protection against damage or deterioration during shipment and allow complete identification of both the filters and the shipping container.

6.1 Each Filter must be marked with the following:
- 6.1.1 The name or trade name of the product/manufacturer
- 6.1.2 The biocide name on the filter

6.2 Shipping containers shall be marked with the following:
- 6.2.1 The name or trade name of the product
- 6.2.2 The filter sizes
- 6.2.3 The carton quantity
- 6.2.4 The manufacturer part number
- 6.2.5 Underwriters' Laboratories, Inc. official listing mark including notation of class.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS
DETAILED BID SPECIFICATIONS

SECTION THREE
HVAC MINI-PLEAT 4V-CELL TYPE
FILTER U.L. CLASS 2

1.0 GENERAL

1.1 Air filters shall be high-efficiency ASHRAE pleat-in-pleat V-bank disposable type assembled in a compact and secure enclosing frame.

1.2 Sizes shall be as noted on drawings or other supporting materials.

2.0 CONSTRUCTION

2.1 Filter media shall be of micro fine glass formed into uniformly spaced pleats separated by hot melt separators and formed into a mini-pleat pack design.

2.2 Each mini-pleat pack shall be assembled into a V-bank configuration with an appropriate number of packs to obtain required pressure drop.

2.3 The media packs shall be bonded to the inside periphery of the enclosing frame with a fire-retardant phosphorus-free sealant.

2.4 The enclosing frame should be comprised of ABS plastic.

2.5 Rigid plastic end caps shall be mechanically fastened to the top and bottom of the enclosing structure to ensure a rigid and durable filter.

2.6 Front face construction shall consist of a ¼" to 1" center point space between the media pack and front face of filter frame, thus minimizing air turbulence and pressure drop growth between pleat pre-filter and final filter.

3.0 PERFORMANCE

3.1 The filter shall have a Minimum Efficiency Reporting Value of MERV 13 when evaluated under the guidelines of ASHRAE Standard 52.2. It shall have a MERV-A of 13 when tested using appendix J of that standard.

3.2 Initial resistance to airflow shall be 0.27".64" w.g at an airflow of 500 fpm.

3.3 Filter shall be rated by Underwriters Laboratories as UL Class 900.

3.4 Manufacturer shall provide evidence of facility certification to ISO 9001:2008.

4.0 SUPPORTING DATA - Provide ASHRAE product test report per ASHRAE Standard 52.2 including testing per appendix J.

4.1 NOMINAL FILTER SIZE (WxHxD)	24x24x12
Nominal Average Dust Spot Efficiency (%)	90-95%
Gross Media Area	200
4.2 TEST CONDITIONS	
Filter Face Velocity (FPM)	492
Test Air Flow Rate (CFM)	1968
Maximum Initial Resistance (Inches W.G.)	0.27"
Final Resistance (Inches W.G.)	1.5
Dust Holding Capacity under 52.2-2012	350 grams
4.3 PERFORMANCE REQUIREMENTS	
Minimum Composite Average Efficiencies (%)	E1: 69; E2: 92; E3: 99
ASHRAE STANDARD 52.2-2012 with Appendix J	
Minimum Efficiency Reporting Value (MERV-A)	13/A

5.0 BID ENCLOSURES

The following items are required as part of the bid response.

4.1 For the master final filter size of 24X24X12, there must be one (1) independent laboratory test report showing performance results when tested to ASHRAE standard 52.2-2012 MERV 13/13A. Permission shall be granted by the bidder to verbally verify the performance results with the independent laboratory that performed the tests.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS
DETAILED BID SPECIFICATIONS

SECTION FOUR
HIGH-EFFICIENCY ASHRAE GRADE POCKET FILTER

1.0 GENERAL

1.1 Air filters shall be high-efficiency, extended surface pocket-style filters consisting of high-loft air-laid micro-fine glass media, a reinforced ABS plastic header, ABS plastic pocket retainers, and bonding agents to prevent air bypass and ensure leak-free performance.

1.2 Sizes shall be as noted on drawings or other supporting materials.

2.0 CONSTRUCTION

2.1 Filter media shall consist of high-density air-laid lofted micro-fine glass media that is chemically bonded to a synthetic micro-mesh media support backing forming a lofted filter blanket.

2.2 Individual pockets shall contain a minimum of 40 stitching support points per square foot of media area. The sides and ends of each pocket shall be sewn with a chain-link overlock stitch.

2.3 Pockets shall be formed into tapered pleats, supported by controlled media space stitching, to promote uniform airflow across the surface of the media. At any point, the sizes of the upstream and downstream passages shall be proportional to the volume of filtered air. The pockets shall also have a conical configuration to minimize contact with HVAC system components.

2.4 Support members shall include an ABS plastic header and ABS plastic pocket retainers. The header shall be joined to the media to prevent air bypass. Individual pocket retainers shall be attached to the header frame with anchor ports allowing for visual confirmation. Bypass between pockets shall be eliminated through a snap-to-seal pocket retainer that shall be an integral part of the two-piece header design. The frame shall form a rigid and durable support assembly.

2.5 The air exiting side of the air tunnels include a pocket flange to ensure pocket integrity throughout the life of the filter. A downstream pocket-to-pocket partition shall provide additional pocket separation to ensure full flow through the entire media area.

2.6 A filter-to-filter sealing gasket shall be installed on one of the vertical members of the filter header.

3.0 PERFORMANCE

3.1 The filter shall have a Minimum Efficiency Reporting Value of MERV (11, 13, 14, 16) when evaluated under the guidelines of ASHRAE Standard 52.2-2012. It shall also have a MERV-A rating of (11, 13, 14, 15) when evaluated under ASHRAE Standard 52.2, Appendix J. It shall have an efficiency of (ePM10-70, ePM1-60, ePM1-70, ePM1-80) when evaluated per ISO filter testing standard 16890.

3.2 Initial resistance to airflow as listed by the manufacturer on a 22" depth 10-pocket bag shall be (0.24", 0.38", 0.45", 0.62")* w.g at an airflow of 500 fpm. Additional information shall be as noted on drawings or other supporting materials.

3.3 The manufacturer shall warranty that the filter shall be capable of withstanding 10.0" w.g. without failure of the filter.

3.4 The filter shall be classified by Underwriters Laboratories as UL 900.

3.5 Manufacturer shall provide evidence of facility certification to ISO 9001:2015.

Supporting Data – Provide product test reports for each listed efficiency, including all details as prescribed in
ASHRAE Standards 52.2 and ISO Standard 16890

4.0 SUPPORTING DATA – Provide ASHRAE product test report per ASHRAE Standard 52.2, including testing per appendix J.

5.0 HIGH EFFICIENCY BAG FILTER SUPPORTING DATA – Provide ASHRAE product test report per ASHRAE Standard 52.2 including testing per appendix J.

4.1 NOMINAL FILTER SIZE (WxHxD) 24x24x22	
Nominal Average Dust Spot Efficiency (%)	80-85%
Gross Media Area	71
4.2 TEST CONDITIONS	
Filter Face Velocity (FPM)	493
Test Air Flow Rate (CFM)	1968
Maximum Initial Resistance (Inches W.G.)	0.37”
Final Resistance (Inches W.G.)	1.5
Dust Holding Capacity under 52.2-2012	444 grams
4.3 PERFORMANCE REQUIREMENTS	
Minimum Composite Average Efficiencies (%)	E1: 71-A; E2: 97-A E3: 100-A
ASHRAE STANDARD 52.2-2012 with Appendix J	
Minimum Efficiency Reporting Value (MERV-A)	13/A

6.0 BID ENCLOSURES

The following items are required as part of the bid response.

5.1 For the master final filter size of 24X24X22, there must be one (1) independent laboratory test report showing performance results when tested to ASHRAE standard 52.2-2012 MERV 13/13A. Permission shall be granted by the bidder to verbally verify the performance results with the independent laboratory that performed the tests.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS
DETAILED BID SPECIFICATIONS

SECTION FIVE
HVAC BELTS

1) All belts must be manufactured in the USA.

2) All orders must be packaged and labeled per facility and per equipment number (no bulk packaging).

3) All orders are to be set up for auto ship per Douglas County's Maintenance Schedule.

4) No minimum order requirements.

5) The belt list is provided, attached, and incorporated herein as Exhibit A, and includes estimated annual quantities.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS
DETAILED BID SPECIFICATIONS

SECTION SIX
CELdek – DIRECT EVAPORATIVE COOLING/HUMIDIFIER MEDIA

1) The media shall be CELdek® cellulose evaporative cooling/humidification media as manufactured by the Munters Corporation. The engineered media shall be self-supporting, impregnated, and treated to give high absorbency and protection against decomposition through the effects of water and air. It shall consist of unique design characteristics to assure even air flow and higher evaporative cooling efficiency while preventing clogging and biological degradation. It shall be a cross-fluted, self-cleaning design. Saturation efficiency for a 12" depth in the direction of the air flow will not be less than 90% when operated at up to 400 feet per minute (fpm) face velocity. The media air passage flutes shall form an angle of 15 degrees with the horizontal air flow, and the water passage flutes shall form an angle of 45 degrees with the horizontal air flow. The depth of the media shall be [6", 8", 12", 18"] in the direction of the air flow.

2) A water distribution pad with a thickness of [2", 3"] shall be provided and installed across the top cross section of the media to provide for the even distribution of water. The water distribution pad shall be constructed of the same cellulose material as that of the cooling / humidification media. The distribution pad shall have a synthetic edge coating applied on the water inlet and outlet sides to reduce erosion and provide for cleaning.

3) The CELdek media shall have an average design life span of between 3 and 5 years when maintained according to the manufacturers recommendations and used during a typical cooling season of 5 to 6 months per year.

4) Media cooling effectiveness and resistance to air flow shall be certified by the manufacturer that it is tested according to ASHRAE 41.2 - 1987 *Standard Methods for Laboratory Air Flow Measurement,* Figure 15.

5) The manufacturer of the media shall be certified to the quality standard system ISO 9001, 1994 at all locations or their operations.

6) The manufacturer shall warrant that the media shall be free from defects in materials and workmanship for a period of one (1) year commencing from the date of shipment. This warranty is provided that the media is installed and maintained according to the manufacturers written instructions and details contained in their Engineering Bulletins.

7) Media manufactured from aspen wood, horsehair, glass wool or any other random or synthetic materials shall not be allowed.

8) Alternate manufacturers shall not be allowed to provide proposals until they have provided to the owner's satisfaction (or their delegated project architect and/or engineer) that the performance of their proposed product is equal to the base bid. Alternate manufacturers shall provide certified documentation and warrant that their product will not cause any objectionable odors, lose rigidity, thermal efficiency, deteriorate or in any way fail prematurely.

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS

HVAC MERV 9 PLEAT FILTERS
BID SCHEDULE

TYPE: HVAC MERV 9 PLEAT FILTERS						
Size	**Estimated Annual Usage**	**Minimum Order Quantity**	**Volume Price Break**	**Unit Price**	**Manufacturer**	**Model Number**
16x16x2	8					
16x20x2	295					
16x25x2	162					
20x20x2	245					
20x24x2	133					
20x25x2	265					
25x16x2	245					
25x20x2	356					
25x25x2	10					
24x12x2	36					
24x16x2	115					
24x20x2	118					
24x24x2	56					
13.5x19.5x2	4					
34.5x15.5x2	2					
34.5x24.5x2	2					
20x16x4	16					
20x20x4	12					
20x24x4	32					
24x12x4	12					
24x24x4	40					
25x16x4	66					
25x20x4	178					

Performance Criteria	
ECI Value of 5 Stars	YES / NO
Initial Resistance (in W.G.)	
Final Resistance (in W.G.)	
Minimum Composite Average Efficiency - % E1	
Minimum Composite Average Efficiency - % E2	
Minimum Composite Average Efficiency - % E3	
Merv – Rating (Minimum 9A)	YES / NO
Integrity guaranteed to 2” W.G.	YES / NO
Underwriters Laboratories as UL Class 2	YES / NO

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS

HVAC MERV 9 PLEAT FILTERS
BID SCHEDULE - Continued

Physical Characteristics	
Beverage Board - 28 Point high wet strength	YES / NO
Frame Bonded to Media on All Sides	YES / NO
Media Type (i.e., synthetic, cotton)	
Media Thickness (0.15")	YES / NO

COMMENTS:

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS & BELTS

HVAC RING PANEL 2 PLY
BID SCHEDULE

TYPE: HVAC RING PANEL 2 PLY						
Size	**Estimated Annual Usage**	**Minimum Order Quantity**	**Volume Price Break**	**Unit Price**	**Manufacturer**	**Model Number**
POLY WIRE FRAME						
11-3/4x47x1	4					
9x24x1	12					
12x13x1	120					
12x7x1	12					
15x13x1	12					
15x17x1	120					
19x30x1	72					
19x19x1	24					
11-3/4x54x1	4					
9x9x1	12					
6x60x1	12					
16x28x1	12					
27x27x1	12					
17x24x1	24					
POLY PANEL						
11-3/4x47x1	4					

Performance Criteria	**2 PLY**
Test Air Flow Rate CFM	
Initial Resistance (in W.G.)	
Final Resistance (in W.G.)	
Merv – Rating 7 ASHRARE 52.2 – 1999	N/A
Anti-Microbial Biocide (NOT BIOSTATE)	YES/NO

Physical Characteristics	**2 PLY**
Media Type	
Wire Frame Gauge	
Filter Has Internal Stabilizer Seal	YES / NO
Heat Sealed Around Internal Wire Grid	YES / NO

COMMENTS:

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS and BELTS

MINI-PLEAT 4V-CELL TYPE
FILTER U.L. CLASS 2
BID SCHEDULE

TYPE: MINI-PLEAT 4V CELL TYPE FILTER						
Size	**Estimated Annual Usage**	**Minimum Order Quantity**	**Volume Price Break**	**Unit Price**	**Manufacturer**	**Model Number**
10P MERV 13A - HI-FLO-ES BAG						
20x24x22	4					
24x24x22	24					
10P MERV 14A - HI-FLO-ES BAG						
12x24x22	8					
20x24x22	19					
DF MERV 13A – 4/5-V MINI-PLEAT						
24x12x12	13					
20x20x12	52					
24x20x12	112					
24x24x12	72					
DF MERV 14A – 4/5-V MINI-PLEAT						
24x12x12	12					
24x24x12	16					

COMMENTS:

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
3L180	12				
3L190	6				
3L220	4				
3L270	1				
3L280	2				
3L300	3				
3L370	1				
3L380	1				
3L390	6				
3VX300	2				
3VX530	4				
3VX600	14				
3VX670	8				
3VX900	4				
4L200	12				
4L210	10				
4L220	4				
4L230	1				
4L240	1				
4L250	4				
4L290	8				
4L300	4				
4L310	4				
4L320	3				
4L420	1				
4L590	1				
4L670	2				
4L800	8				

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE - Continued

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
5VX1000	12				
5VX1030	4				
5VX1120	8				
5VX450	4				
5VX490	2				
5VX510	6				
5VX530	2				
5VX550	2				
5VX630	4				
5VX660	2				
5VX670	6				
5VX710	2				
5VX780	2				
5VX900	2				
5VX960	4				
A23	2				
A26	3				
A29	1				
A32	8				
A33	8				
A37	4				
A38	5				
A39	3				
A41	4				
A42	1				
A44	3				
A45	7				
A48	3				
A49	3				
A50	1				
A51	3				

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE – Continued

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
A55	11				
A60	4				
A61	8				
A65	2				
A92	2				
AX21	12				
AX22	11				
AX23	12				
AX24	2				
AX25	1				
AX26	4				
AX27	2				
AX29	4				
AX30	6				
AX31	2				
AX32	2				
AX34	2				
AX35	4				
AX36	4				
AX37	4				
AX38	8				
AX39	2				
AX40	2				
AX41	4				
AX42	17				
AX45	4				
AX51	4				
AX53	8				
AX54	4				
AX55	8				
AX56	8				

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE – Continued

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
AX58	2				
AX59	2				
AX61	4				
AX62	4				
AX64	8				
AX67	2				
AX68	12				
AX69	14				
AX70	8				
AX71	8				
AX96	8				
B112	4				
B70	6				
B73	3				
B78	2				
B92	6				
B93	4				
BX103	6				
BX35	2				
BX38	18				
BX40	4				
BX41	2				
BX42	8				
BX45	2				
BX46	1				
BX47	2				
BX48	4				
BX50	4				
BX59	4				
BX61	6				
BX62	2				
BX63	3				

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE – Continued

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
BX66	5				
BX68	4				
BX70	7				
BX71	2				
BX74	3				
BX76	2				
BX77	2				
BX78	2				
BX80	9				
BX81	2				
BX82	2				
BX85	3				
BX93	2				
BX95	2				
M411425P	2				

COMMENTS:

INVITATION FOR BID (IFB) #042-26
HVAC FILTERS and BELTS

CELdek – Direct Evaporative Cooling/Humidifier Media
BID SCHEDULE

TYPE: CELdek – Direct Evaporative Cooling/Humidifier Media			
Estimated Usage	**Unit Price Per Cubic Foot**	**Manufacturer**	**Model Number**
214 cubic ft			

COMMENTS: